15 Linear Transformation visualization

Say I have a Transformation (T)

$T: \mathbb{R} \rightarrow \mathbb{R}$ 1 Dimension

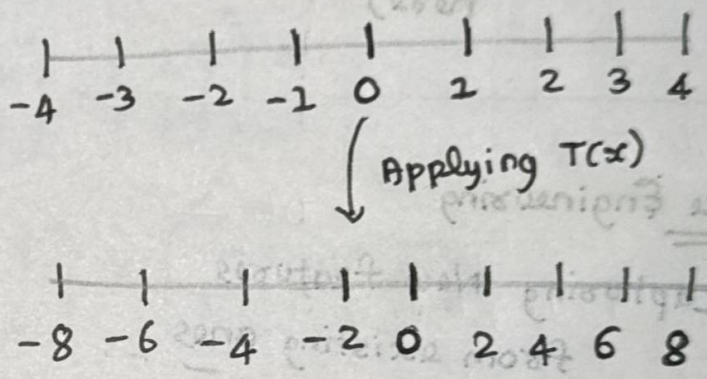
(maps from real no. $\rightarrow$ real no.)

I have a fn.

$\boxed{T(x) = 2x}$

They are in the same linear line

Reason why we say it follows linear Trans



Imp property (linear Trans)

1. Origin (0) $\rightarrow$ must be fixed
2. All lines must remain lines

16 Magnitude and Unit Vector

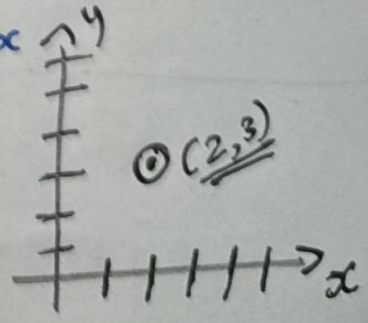
Vector length

$\vec{x} \in \mathbb{R}^n$

vector — belong to Real no. (n-dimension)

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Unit vector $\rightarrow$ vector has a length of 1

$$A = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

x, y